

TestLab 29119 — User Manual

Course: SENG 436 — Learner-as-Designer Project
Institution: Çankaya University
Standards: ISO/IEC 29119-4 (Test Techniques) · ISO/IEC 25010 (Quality Model)
Version: 1.0 · May 2026

1. Introduction

TestLab 29119 is a browser-based educational serious game. You play the role of a **Test Manager** who must classify, apply, and select ISO/IEC 29119-4 test design techniques for realistic software projects. The game reinforces correct technique choice—not memorization of definitions alone—through four progressive phases and immediate feedback.

2. System Requirements

Requirement	Details
Browser	Recent Chrome, Microsoft Edge, or Firefox
JavaScript	Must be enabled
Internet	Required for Google Fonts (optional fallback to system fonts)
Storage	<code>localStorage</code> used for best score, leaderboard, and theme
Installation	None (web application)

3. How to Access the Game

Online (recommended for submission)

Live URL: <https://ismaiidogan.github.io/TestLab-29119/>

Local (development or offline demo)

```
npm install
npm start
```

Open `http://localhost:3000` in your browser.

4. Game Overview

Item	Description
Maximum score	500 points
Phases	4 + Results
Duration	Approximately 25–40 minutes (first playthrough)
Language	English

Phase summary

Phase	Name	Max points	Skill tested
1	Classify	110	Map techniques to Specification / Structure / Experience categories
2	Apply	195	Choose techniques for nine realistic scenarios
3	Strategize	110	Build a technique mix for three projects (Clause 4 constraints)
4	Test Cases	85	Select required EP and BVA test cases
—	Results	—	Grade, misconceptions, certificate, leaderboard

5. Getting Started

5.1 Welcome screen

1. Optionally enter **Your Name** and **School Name** (used on the leaderboard and certificate).
2. Click **Start Mission** to open the tutorial.
3. Defaults: Anonymous / Unknown School if fields are left empty.

5.2 Tutorial (six slides)

Slide	Topic
1	Welcome — Test Manager role
2	ISO/IEC 29119-4 — three technique categories
3	The three categories (Specification / Structure / Experience)
4	Clause 4 selection criteria
5	ISO/IEC 25010 link to Phase 2 scenarios

Slide	Topic
6	Your mission — overview of all four phases

- Use **Next** / **Back** and the dot indicator to navigate.
- Click **Skip** to go directly to Phase 1.
- On the last slide, **Start Phase 1** begins the game.

6. Phase-by-Phase Instructions

Phase 1 — Classify the Techniques

Goal: Drag each of **15 technique cards** into the correct category drop zone.

Zone	Category
Blue	Specification-Based
Green	Structure-Based
Red	Experience-Based

Actions:

- **Drag and drop** cards from the pool into a zone.
- **i** button — opens ISO detail modal (clause, description, best use).
- **Hint** button — reveals a hint; costs **4 points** per first use on each card.
- When all cards are placed, click **Check Answers**.
- Correct placements are highlighted in green; wrong cards show feedback with the correct category.
- Click **Continue to Phase 2** when finished.

Scoring: 6 points per correct card; +20 bonus if all 15 are correct; hint penalties subtracted from Phase 1 total (max 110).

Phase 2 — Apply to Scenarios

Goal: Answer **9 multiple-choice scenarios** by choosing the best test technique(s).

Features:

- Each scenario shows context, a question, and four options (A–D).
- **Keyboard:** Press **A**, **B**, **C**, or **D** to select; **Escape** closes modals.
- **ISO/IEC 25010 badge** on the scenario header links quality characteristics to the scenario.
- Some scenarios are **comparison mode** (e.g. EP vs BVA, Exploratory vs Ad Hoc).

- After answering, read **correct/wrong feedback** and any **misconception** note.
- Click **Next Scenario** until all nine are complete.

Scoring: 20–25 points per scenario (max 195 for Phase 2).

Phase 3 — Build Your Test Strategy

Goal: For each of **three projects**, select up to **five** test techniques that best fit the constraints.

Projects:

1. MedSafe — Healthcare (safety-critical)
2. QuickMVP — Startup food delivery (time/budget limited)
3. MigrateX — Legacy ERP migration (undocumented rules)

Actions:

- Read **constraints** (risk, budget, timeline, team, documentation).
- Click techniques to select/deselect (maximum 5).
- Hover for a quick preview; use the glossary (book icon) for ISO terms.
- Click **Submit Strategy** to see score, optimal selection, expert analysis, and key insight.
- Click **Next Project** or **Continue to Phase 4** on the last project.

Scoring: Up to 37 / 37 / 36 points per project (max 110). Partial credit applies; there is no single fixed algorithm—matching Clause 4 professional judgment.

Phase 4 — Write Your Test Cases

Goal: For **two exercises**, select all **required** EP and BVA test cases (checkboxes).

Exercises:

1. Login form — username field (5–20 characters)
2. Age field — insurance discount (18–65 inclusive)

Actions:

- Click rows to toggle selection.
- Click **Submit** to see coverage feedback (correct, missed, or unnecessary selections).
- Extra (non-required) selections incur a small penalty.
- Click **Next Exercise** or **See Final Results** on the last exercise.

Scoring: Up to 42 and 43 points (max 85). Each required item = 6 points when selected.

7. Scoring and Grades

Total score

Your **total** is the sum of all four phase scores (max **500**). The navbar **Score** updates as you play; **Best** stores your highest total in this browser.

Grade tiers (percentage of 500)

Grade	Minimum %	Title
ISO Master	90%	Outstanding mastery
Senior Tester	75%	Strong understanding
Test Engineer	60%	Good basics
Junior Tester	40%	Learning in progress
Trainee	0%	Review recommended

8. User Interface Reference

Navbar (top)

Control	Function
Top 10	Local leaderboard (this browser only)
Best	Highest total score saved locally
Score	Current run total
Book	ISO reference glossary (searchable)
Sun/Moon	Light / dark theme
Speaker	Mute / unmute sound effects

Phase progress bar

Shows four steps: **Classify** → **Apply** → **Strategize** → **Test Cases**. Active and completed steps are highlighted during gameplay.

9. Results Screen

After Phase 4 you will see:

- **Grade title** and emoji (e.g. ISO Master)
 - **Total points** and percentage
 - **Animated progress bar**
 - **Per-phase score breakdown**
 - **Misconceptions encountered** (learning points from wrong answers)
 - **Phase 3 key insights**
 - **Hint usage** summary (Phase 1)
 - **Download Certificate** — PNG file with your name, grade, and scores
 - **Play Again** — returns to welcome (scores reset for a new run)
-

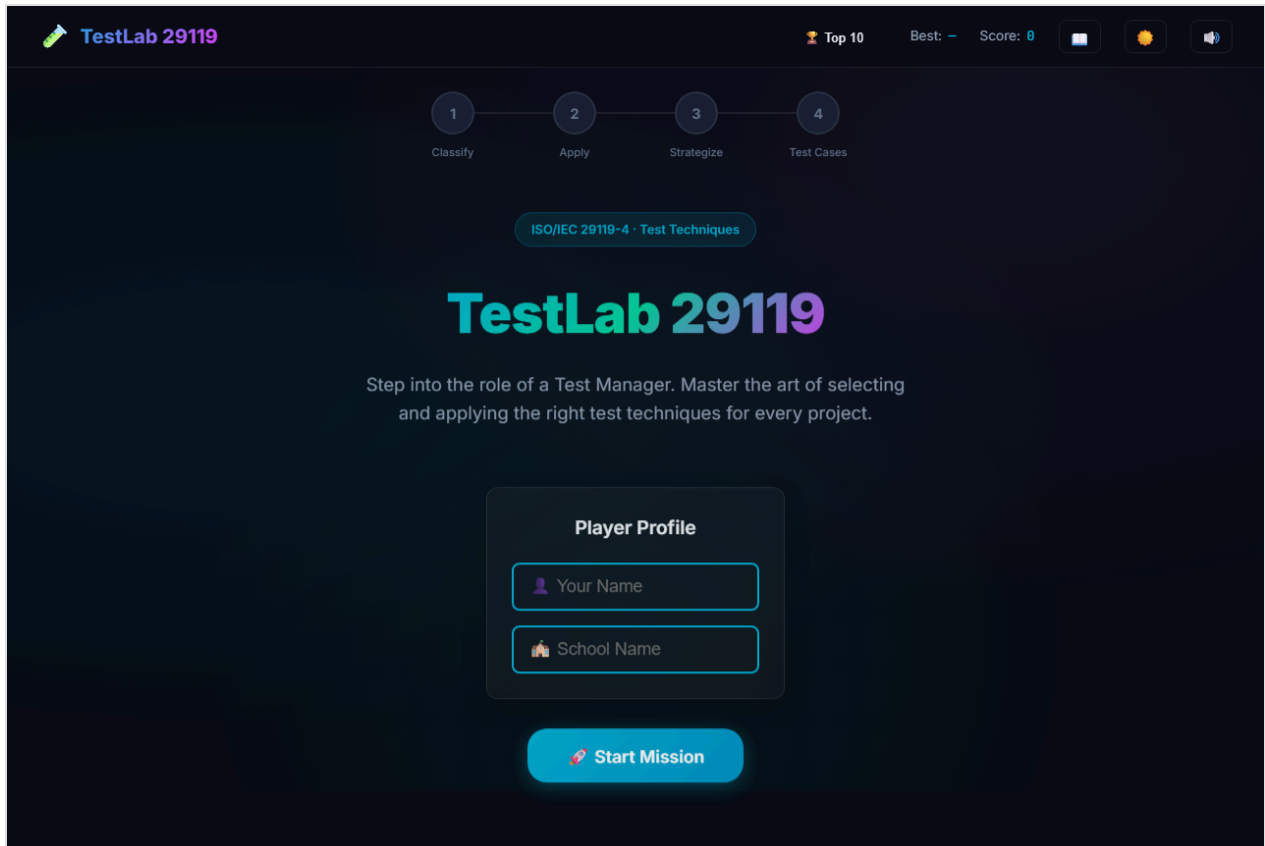
10. Troubleshooting

Problem	Solution
Score or leaderboard wrong	Data is per-browser; clear site data or use another browser
Fonts look plain	Check internet connection (Google Fonts CDN)
Drag-and-drop not working	Use a desktop browser; refresh the page
Game does not load	Ensure JavaScript is enabled; use the GitHub Pages URL or <code>npm start</code>

11. Appendix A — Screenshot Walkthrough

TestLab 29119 is a web application accessed via URL. The following screens document the user interface in play order.

1. Welcome — Player profile



The screenshot shows the 'TestLab 29119' player profile interface. At the top, there's a header with the logo, a 'Top 10' indicator, and a 'Score: 0' display. Below the header is a progress bar with four steps: 1. Classify, 2. Apply, 3. Strategize, and 4. Test Cases. A central banner reads 'ISO/IEC 29119-4 · Test Techniques' and 'TestLab 29119'. Below this, a description states: 'Step into the role of a Test Manager. Master the art of selecting and applying the right test techniques for every project.' The 'Player Profile' section contains two input fields: 'Your Name' and 'School Name'. At the bottom is a prominent 'Start Mission' button.

TestLab 29119

Top 10 Best: — Score: 0

1 2 3 4
Classify Apply Strategize Test Cases

ISO/IEC 29119-4 · Test Techniques

TestLab 29119

Step into the role of a Test Manager. Master the art of selecting and applying the right test techniques for every project.

Player Profile

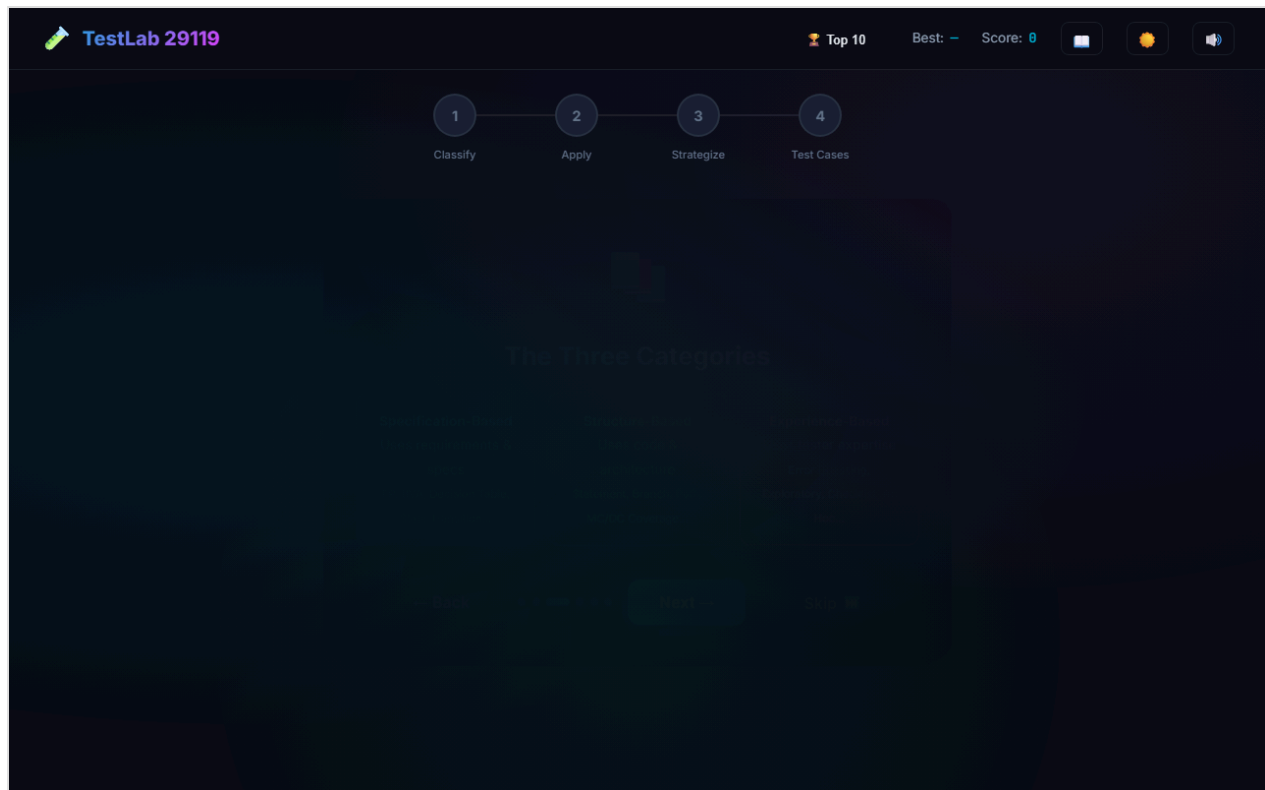
 Your Name

 School Name

 **Start Mission**

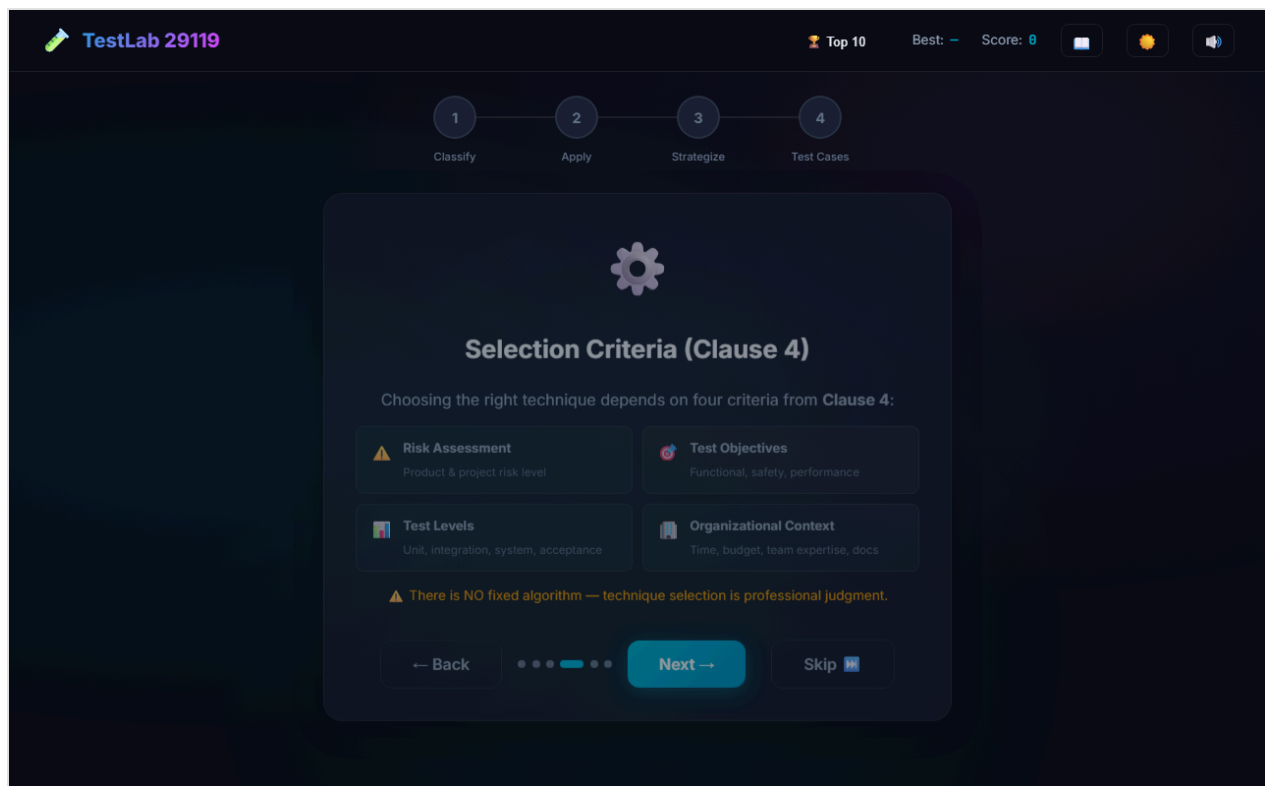
Enter name and school, then **Start Mission**. Profile data appears on the certificate and leaderboard.

2. Tutorial — Three technique categories



Introduces Specification-Based, Structure-Based, and Experience-Based groups (ISO/IEC 29119-4).

3. Tutorial — Clause 4 selection criteria



Risk, objectives, test levels, and organizational context—the foundation for Phase 3.

4. Phase 1 — Classify

The screenshot displays the TestLab 29119 interface. At the top, a progress bar shows four steps: 1. Classify (active), 2. Apply, 3. Strategize, and 4. Test Cases. Below the progress bar, the title "PHASE 1 Classify the Techniques" is centered. A subtitle reads "Drag each test technique card to its correct category. 0 / 15 placed".

Below the subtitle, there is a grid of 15 test technique cards, each with a title and a small icon. The cards are:

- Statement Coverage
- Use Case Testing
- State Transition Testing
- Error Guessing
- Path Coverage
- Branch Coverage
- Checklist-Based Testing
- Pairwise Testing
- Exploratory Testing
- MC/DC Coverage
- Ad Hoc Testing
- Classification Tree Method
- Decision Table Testing
- Boundary Value Analysis
- Equivalence Partitioning

At the bottom of the interface, there are three category drop zones, each with a colored square icon and a label:

- Specification-Based (blue square)
- Structure-Based (green square)
- Experience-Based (red square)

Drag fifteen technique cards into the three category drop zones.

5. Phase 2 — Scenario with ISO 25010 badge

The screenshot displays the TestLab 29119 interface. At the top, a progress bar shows four steps: 1. Classify (highlighted), 2. Apply, 3. Strategize, and 4. Test Cases. The main heading is "Apply to Scenarios". Below this, a scenario titled "Login Form Validation" is presented. The scenario text states: "You are testing a login form that accepts a username (6-20 characters) and password (8-30 characters). The system should reject inputs outside these ranges and accept valid ones." The question asks: "Which TWO techniques should you apply together for maximum effectiveness?". Four options are listed: A. Equivalence Partitioning + Boundary Value Analysis, B. Statement Coverage + Branch Coverage, C. Error Guessing + Ad Hoc Testing, and D. Path Coverage + MC/DC Coverage. The interface also shows a "Top 10" badge, a "Best" score of 0, and a "Score" of 0.

Nine multiple-choice scenarios; each shows an ISO/IEC 25010 quality characteristic badge.

6. Phase 3 — Test strategy

 **TestLab 29119**

Top 10Best: —Score: 0

1Classify

2Apply

3Strategize

4Test Cases

PHASE 3

Build Your Test Strategy

Given a project with real constraints, select the optimal technique mix. Project 1 / 3

 **MedSafe — Healthcare Patient Management System**

A hospital is deploying a new patient management system that handles medication dosages, patient records, and appointment scheduling. The system is safety-critical — incorrect dosage calculations could harm patients.

**RISK ASSESSMENT**
RISK
Very High (safety-critical, lives at stake)

**ORGANIZATIONAL CONTEXT**
BUDGET
High (hospital budget allocated)

**ORGANIZATIONAL CONTEXT**
TIMELINE
12 weeks for testing

**ORGANIZATIONAL CONTEXT**
TEAM
5 experienced testers, 2 with medical domain knowledge

**ORGANIZATIONAL CONTEXT**
DOCUMENTATION
Complete requirements specification available

 **Available Techniques**

- Equivalence Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Statement Coverage
- Branch Coverage
- MC/DC Coverage
- Error Guessing
- Exploratory Testing

Selected: 0 / 5

 **Your Strategy**

Select 5 techniques from the left panel

Submit Strategy

Select up to five techniques per project given risk, budget, timeline, team, and documentation constraints.

7. Phase 4 — Test case selection


TestLab 29119


Top 10

Best: —

Score: 8







1

Classify

2

Apply

3

Strategize

4

Test Cases

PHASE 4

Write Your Test Cases

Select test cases you would include for the given scenario. Exercise 1 / 2

EP + BVA

Login Form — Username Field

A username field accepts 5-20 characters. The system must reject values outside this range. Select ALL test cases you would include using Equivalence Partitioning and Boundary Value Analysis.

☐
☒
Equivalence Partitioning 1 (PASS)

☐
☒
Boundary Value Analysis 1 (EXISTS)

☐
☒
Security: SQL Injection

☐
☒
UX: special characters only

☐
☒
EP — Valid partition: 10 chars (e.g. "testuser1")

☐
☒
EP — Invalid partition: 2 chars (below minimum)

☐
☒
EP — Invalid partition: 25 chars (above maximum)

☐
☒
BVA — Boundary: 4 chars (just below min — expect FAIL)

☐
☒
BVA — Boundary: 5 chars (minimum valid — expect PASS)

☐
☒
BVA — Boundary: 20 chars (maximum valid — expect PASS)

☐
☒
BVA — Boundary: 21 chars (just above max — expect FAIL)

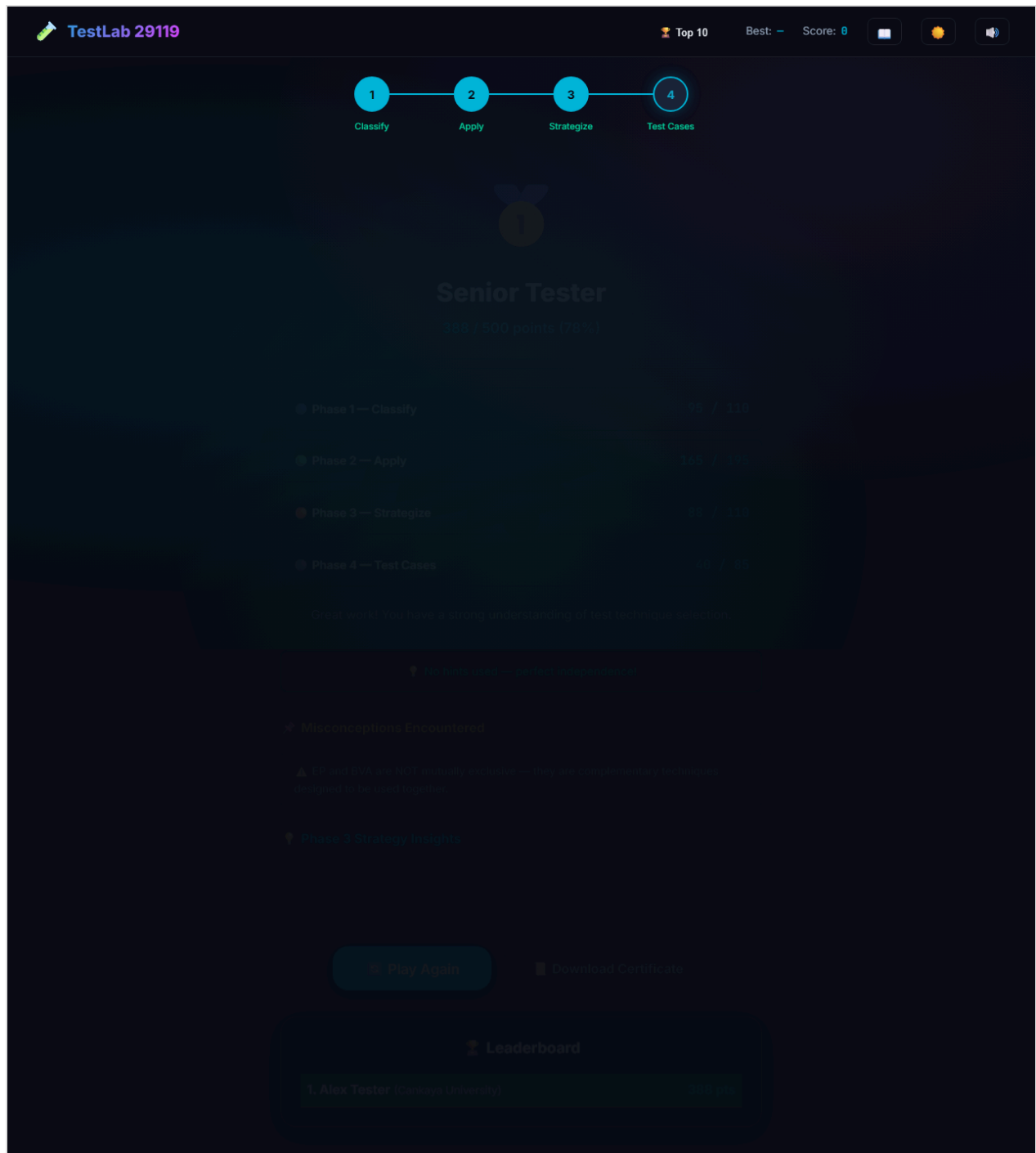
☐
☒
Security: SQL injection input " OR 1=1"

☐
☒
UX: special characters only "QW!\$%&'

✓ Submit Test Plan

Choose required EP and BVA test cases via checkboxes.

8. Results — Score and grade



Total score, grade tier, phase breakdown, misconceptions, and certificate download.

9. ISO reference glossary

TestLab 29119

Top 10 Best: — Score: 0

1 2 3 4
Classify Apply Strategize Test Cases

Senior Tester
388 / 500 points (78%)

Phase 1 — Classify	95 / 110
Phase 2 — Apply	165 / 195
Phase 3 — Strategize	88 / 110
Phase 4 — Test Cases	40 / 85

Great work! You have a strong understanding of test technique selection.

💡 No hints used — perfect independence!

✖ **Misconceptions Encountered**

⚠ EP and BVA are NOT mutually exclusive — they are complementary techniques designed to be used together.

💡 **Phase 3 Strategy Insights**

1 Safety-critical systems demand the most rigorous techniques. MC/DC is often REQUIRED by safety standards, not optional.

[Play Again](#) [Download Certificate](#)

Leaderboard

1. Alex Tester (Cankaya University)	388 pts
-------------------------------------	---------

Searchable glossary available from the navbar book icon during play.

End of User Manual